Title: Discrete Mathematics (core course)

Course code: SC612

Structure: 3-1-0-4

Course contents (tentative):

Mathematical logic & Proof techniques: Proposition logic, predicate logic and quantifiers, logic operators, logic proposition Introduction to different standard proof techniques

Basic discrete structures: Set operations, Functions, Relations and their properties, Equivalence relations, Closure of relations, Partial orderings, Sequences and summations, Matrices

Number Theory: Modular arithmetic, primes and representation of integers, linear congruences

Recurrence relations: Solving linear recurrence relations, Divide and conquer algorithms and recurrence relations, its application to analysis of algorithms
Induction and recursion: Induction, strong induction, well-ordered property, recursion, structural induction, and generalized induction;

Counting: Combinatorial Principles and Techniques, permutations and combinations, binomial coefficients and identities, Pigeon hole principle, inclusion exclusion principle

Discrete Probability: Basic probability theory, Bayes' theorem, expected value and variance

Graph Theory: Undirected graph, Representations, Handshaking lemma, Euler path and circuit, Hamiltonian cycle, Complete graph, cycles, bipartite graph, planar graphs, Euler formula, shortest path algorithm, Traveling salesman problem, trees. properties of trees, spanning trees

Boolean Algebra: Boolean functions, logic gates, minimization expressions, K-maps

Course Outcomes:

- Logic Proofs: Develop the ability to construct and analyze formal logical proofs using propositional and predicate logic.
- Basics of Counting: Master fundamental counting principles including permutations, combinations, and the principle of inclusion-exclusion.
- Mathematical Induction: Understand and apply mathematical induction to prove statements about integers and sequences.
- Recursive Algorithms: Design and analyze recursive algorithms for solving problems such as sorting, searching, and graph traversal.
- Boolean Algebra: Explore Boolean functions, truth tables, Boolean expressions, and their applications in digital circuits and computer science.

- Discrete Probability: Gain proficiency in applying probability theory to discrete scenarios, including basic probability distributions and expected values.

These outcomes encompass essential topics covered in a typical discrete mathematics course, providing students with foundational knowledge and problem-solving skills crucial for further studies in mathematics, computer science, and related fields.

Text book:

- Discrete Mathematics and its Applications - Kenneth.H. Rosen
- Elements of Discrete Mathematics C. L. Liu Publisher: Tata McGrawHill

Grading policy (tentative): The grade earned by a student will depend on (1) the total marks earned by him in all the exams conducted over the entire semester, and (2) active participation in the class discussions and labs. The weightage of each part is as follows:
Insem 1: 20%
Insem 2: 20%
Endsem: 40%
Tutorials/Lecture Participation/ Assignments: 20 %

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
Y	Y	Y									Y